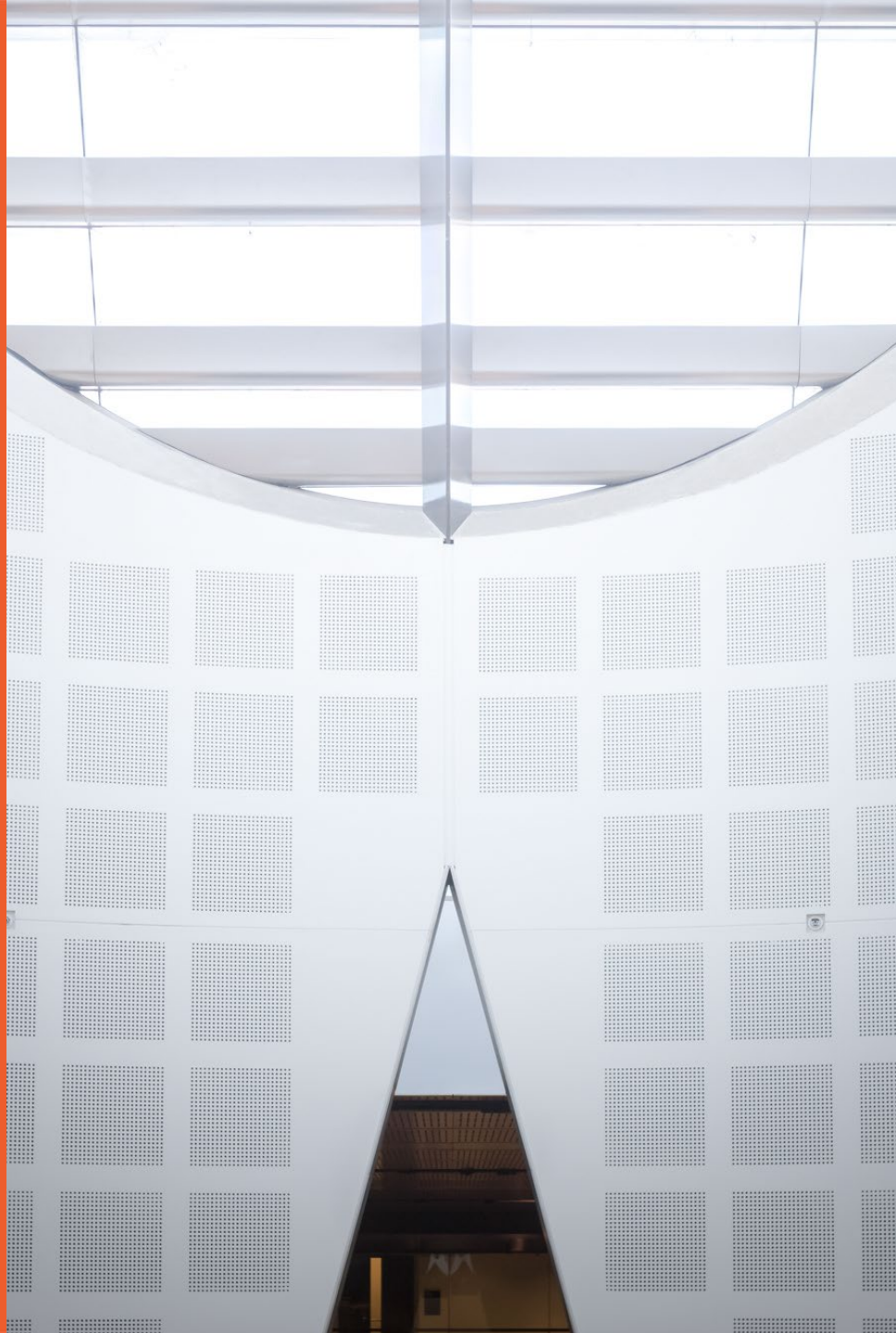
Intro. to matrix algebra & input-output calculations

Lecturer: Dr Amanda Irwin
Unit coordinator: Prof. Arunima Malik
amanda.irwin@sydney.edu.au
arunima.malik@sydney.edu.au

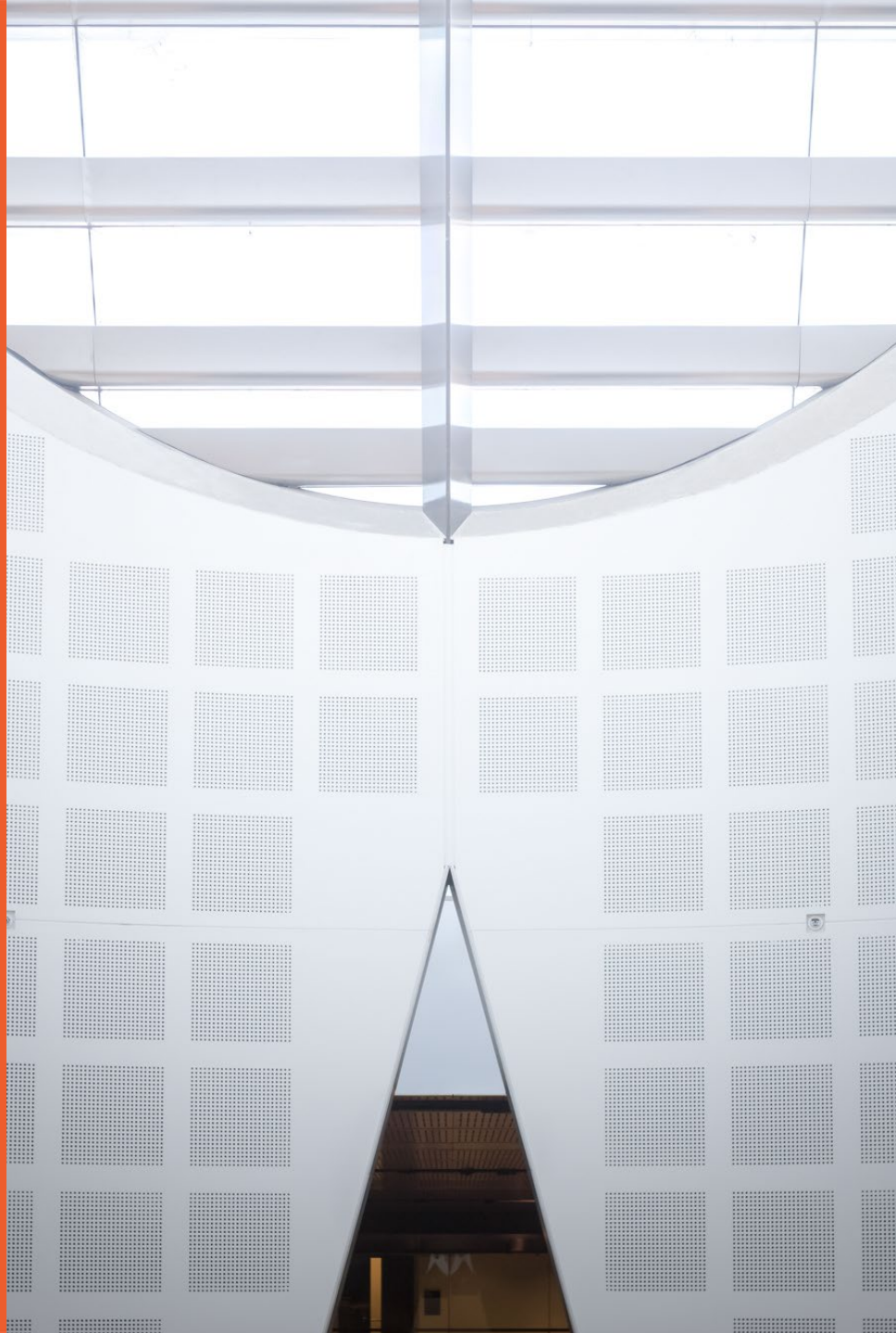
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We recognise and pay respect to the Elders and communities – past, present, and emerging – of the lands that the University of Sydney's campuses stand on. For thousands of years they have shared and exchanged knowledges across innumerable generations for the benefit of all.



PHYS5033



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Recap – Week 1

- What supply chains are
- Why we want to quantify impacts embodied in them
- Some examples of supply chain assessments
- Structure of input-output tables
- Create your own input-output table

Media and PHYS5033 – motivation

The Guardian

News | Opinion | Sport | Culture | Lifestyle

Australia | World | AU politics | Environment | Climate crisis | Indigenous Australia | Immigration | Media | Business | Health | Science | Tech | Podcasts | Newsletters

Australia news

Donald Trump hits Australian exports to US with new higher trade tariff over claims of 'forced labour'

Australia's trade minister, Don Farrell, says tariff regime is unjustified and inconsistent with free trade agreement with US

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https://www.theguardian.com/australia-news/2026/jul/24/donald-trump-hits-australian-exports-to-us-with-new-higher-trade-tariff-over-claims-of-forced-labour

United Nations | UN News
Global perspective Human stories

https://news.un.org/en/story/2026/06/1167658

AI's environmental costs threaten water, land and climate

© Unsplash/Geoffrey Moffett | A data centre in Ireland.

4 June 2026 | Climate and Environment

Artificial Intelligence is not only responsible for worrying amounts of earth-warming greenhouse gases: the technology's environmental footprint is also expanding at a pace that could strain the planet's natural resources.

Data centres, the global infrastructure powering AI, could consume 945 terawatt-hours of electricity annually by 2030 – nearly triple the combined annual electricity use of Pakistan, Bangladesh, and Nigeria, countries collectively home to more than 650 million people.

However, this is just the tip of the iceberg. On top of the carbon footprint, every unit of electricity used by data centres also carries a "water footprint" for cooling and energy production, and a "land footprint" associated with power generation and supply chains.

Rethinking how sustainability is measured

According to a new study from UN University (UNU), **AI-related water consumption could equal the basic annual domestic needs of 1.3 billion people by the end of the decade**, while its **land footprint may exceed 14,500 square kilometres — roughly twice the size of the Jakarta metropolitan area**.

Climate change | + Add to myFT

Activists criticise insurance industry's first attempt to measure carbon emissions

Insurers have been accused by activists of watering down the sector's first attempt to quantify the climate impact of their underwriting portfolios.

Insurance and reinsurance plays a crucial role in the energy sector but companies that provide the cover for oilfields and mines have provided scant details of the emissions they are responsible for.

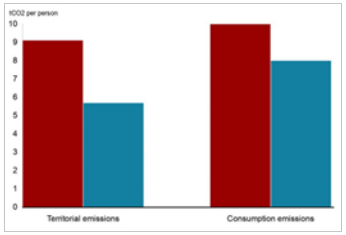
France's Axa and Germany's Allianz are among the insurers preparing to assess the carbon footprint of their underwriting business after the Partnership for Carbon Accounting Financials released standards for the industry in November.

Activists, however, say that the new industry group standard allows insurers to omit the largest chunk of emissions associated with underwriting portfolios — so-called scope 3 emissions made up of greenhouse gases emitted not by the companies themselves but along their supply chains and by those who use their products.

https://www.ft.com/content/628beeaf-1ede-4bd1-8e92-dd70d7636246

Climate change: Does Germany produce double the UK's carbon emissions?

By Nicholas Barrett
BBC News
2 days ago



https://www.bbc.com/news/58148881

You can't manage what you don't measure !

Relevant for individuals, businesses and nations

How and why to measure? Can we identify hot-spots?

Legislation and PHYS5033 - motivation

2022-2023-2024

The Parliament of the Commonwealth of Australia

HOUSE OF REPRESENTATIVES

29

Climate-related metrics

An entity shall disclose information relevant to the cross-industry metric categories of:

(a) *greenhouse gases*—the entity shall:

(i) disclose its absolute gross greenhouse gas emissions generated during the reporting period, expressed as metric tonnes of CO_2 *equivalent* (see paragraphs B19–B22), classified as:

(1) *Scope 1 greenhouse gas emissions*;

(2) *Scope 2 greenhouse gas emissions*; and

(3) *Scope 3 greenhouse gas emissions*;

(ii) measure its greenhouse gas emissions in accordance with the Greenhouse Gas Protocol: A Corporate Accounting and Reporting Standard (2004) unless required by a jurisdictional authority or an exchange on which the entity is listed to use a different method for measuring its greenhouse gas emissions (see paragraphs B23–B25);

(iii) disclose the approach it uses to measure its greenhouse gas emissions (see paragraphs B26–B29) including:

(1) the measurement approach, inputs and assumptions the entity uses to measure its greenhouse gas emissions;

(2) the reason why the entity has chosen the measurement approach, inputs and assumptions it uses to measure its greenhouse gas emissions; and

(3) any changes the entity made to the measurement approach, inputs and assumptions during the reporting period and the reasons for those changes;

https://standards.aasb.gov.au/aasb-s2-sep-2024

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Attorney-General's portfolio

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Albanese Government combatting modern slavery

Home / Media centre / Albanese Government combatting modern slavery

The Hon Michelle Rowland MP

16 July 2026 | Media Release

The Albanese Government is strengthening Australia's modern slavery laws and equipping companies to better identify and address risks in their global supply chains.

Modern slavery is a major violation of human rights and has devastating impacts, depriving victims of their basic freedoms and dignity.

Australia has a robust legislative framework and a world-leading approach to addressing modern slavery, but we can and should do more.

The Government will strengthen Australia's modern slavery laws with a new criminal offence where companies with an annual consolidated revenue over $100 million fail to prevent modern slavery in their supply chains.

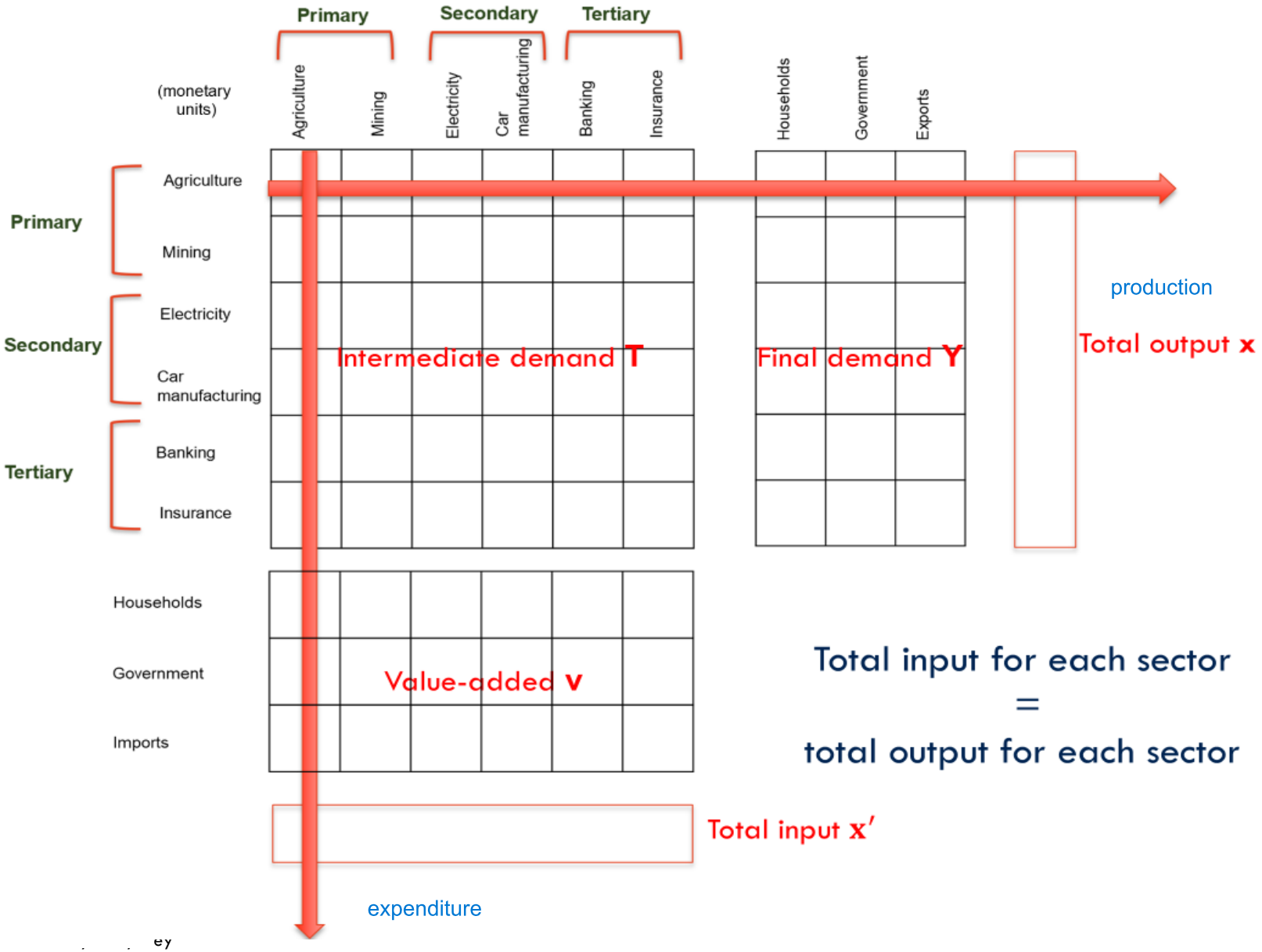
https://ministers.ag.gov.au/media-centre/albanese-government-combatting-modern-slavery-16-07-2026

You can't manage what you don't measure !

Relevant for individuals, businesses and nations

How and why to measure? Can we identify hot-spots?

Recap: Week 1 - Input-output table



MATRIX ALGEBRA



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Key terms

Matrix

A rectangular array of quantities or expressions in rows and columns that is treated as a single entity and manipulated according to particular rules

Vector

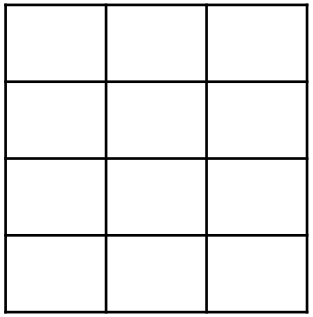
A matrix with one dimension equal to 1 is also known as a vector

Size

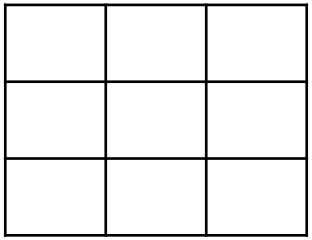
The size of a matrix is expressed in terms of the number of rows and columns, with the number of rows always noted first. For example, a matrix with 3 rows and 4 columns is a 3 x 4 matrix. If the number of rows equals the number of columns, the matrix is said to be square

Elements

The numbers in each 'cell' of the matrix are known as the elements or entries



matrix
size = 4 x 3



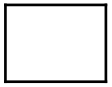
square matrix
size = 3 x 3



row vector
size = 1 x 3



column vector
size = 3 x 1



scalar
(aka a 'normal' number)
size = 1 x 1

Mathematical notation conventions

Matrix size = m x n where:

m = number of rows

n = number of columns

Matrix elements can be identified using algebra - let's call our matrix **A**

$$\mathbf{A}_{mxn} = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}$$

(where '…' indicates lots of numbers that we won't bother including)

Each element in **A** can be identified by its row and column position, so **A** can also be written as

$$\mathbf{A} = (a_{ij}) \text{ where } i = 1,2,\dots,m \text{ and } j = 1,2,\dots,n$$

Mathematical notation conventions

$\mathbf{A}_{mxn} =$

5	10	32	1	81	17
23	0	9	3	7	37
56	1	105	55	65	2
7	-6	4	-4	81	19
27	7	72	48	99	3

What is the value of m? 5

What is the value of n? 6

What is the value of element a_{11}? 5

What is the value of element a_{15}? 81

What is the value of element a_{24}? 3

What is the value of element a_{42}? -6

Matrix properties

Two matrices are equal if, and only if, they are the same size AND have the same corresponding elements.

1	2	3
4	5	6
7	8	9

same size,
elements ≠
∴ not equal

3	2	1
6	5	5
9	8	7

1	2	3
4	5	6
7	8	9

same size,
elements =
∴ equal

1	2	3
4	5	6
7	8	9

Question

Are these matrices equal?

1

2	5	5
3	2	9
5	4	8
6	5	4

2	5	5
3	2	9
5	4	8
6	5	4

2

2	5
3	2
5	4
6	5
6	5

3	2
5	4
6	5
6	5

Matrix operations

Matrix tranposition 轉置矩陣

A matrix is transposed when the rows and columns are interchanged (row 1 becomes column 1 and column 1 becomes row 1 and so on). The transposed matrix is usually denoted with the ’ symbol.

A

1	2	3
4	5	6

A’

1	4
2	5
3	6

B

1
2
3
4

B’

1	2	3	4

Matrix operations

Addition and subtraction

In order to add or subtract matrices, the matrices in question MUST be the same size

1	2	3
4	5	6
7	8	9

+

1	2	3
4	5	6
7	8	9

=

1+1	2+2	3+3
4+4	5+5	6+6
7+7	8+8	9+9

=

2	4	6
8	10	12
14	16	18

Matrix operations

Addition and subtraction

In order to add or subtract matrices, the matrices in question MUST be the same size

1	2
4	5
7	8

-

2	3
5	6
8	9

=

1-2	2-3
4-5	5-6
7-8	8-9

=

-1	-1
-1	-1
-1	-1

Matrix operations

Scalar multiplication

If **A** is any matrix and q is any real number (ie a scalar), the product q**A** is the matrix obtained by multiplying each element of **A** by q

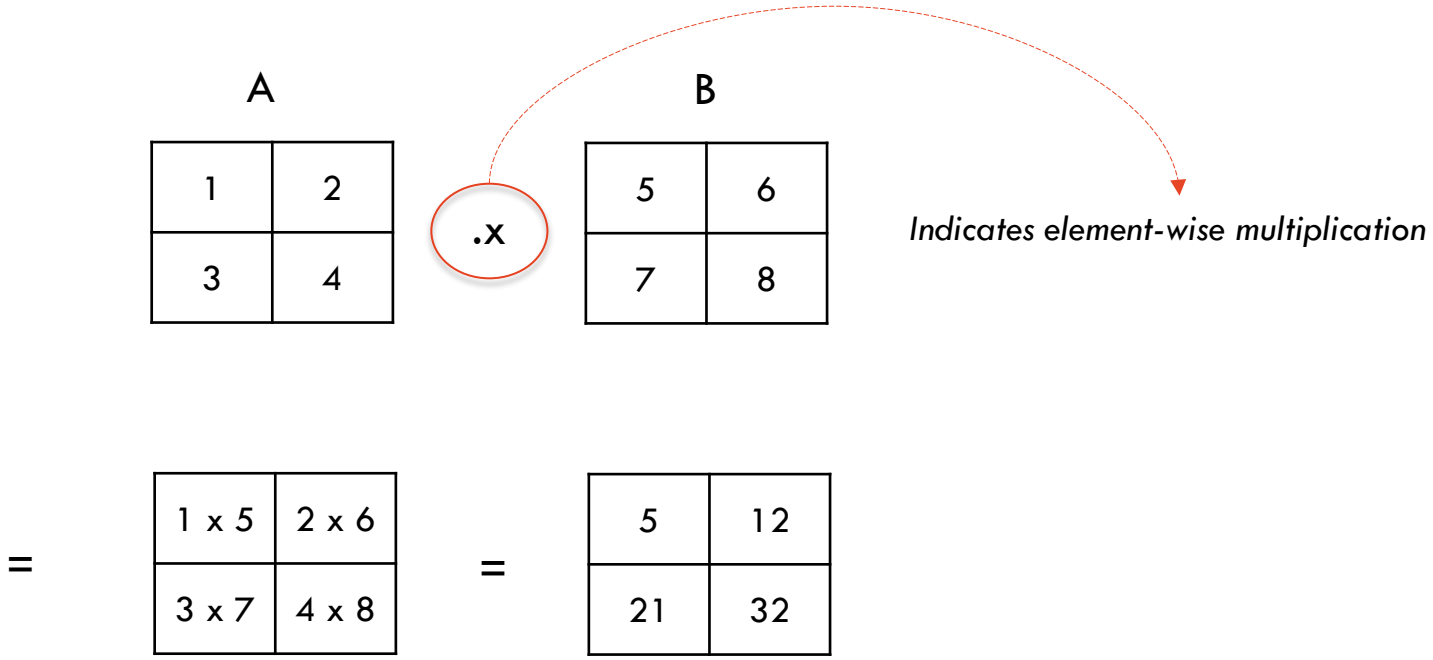
$$3 \times \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} = \begin{bmatrix} 3 \times 1 & 3 \times 2 & 3 \times 3 \\ 3 \times 4 & 3 \times 5 & 3 \times 6 \\ 3 \times 7 & 3 \times 8 & 3 \times 9 \end{bmatrix}$$

$$= \begin{bmatrix} 3 & 6 & 9 \\ 12 & 15 & 18 \\ 21 & 24 & 27 \end{bmatrix}$$

Matrix operations

Element-wise multiplication 逐元素乘法

If two matrices are the same size, we can carry out element-wise multiplication



Matrix operations

Matrix multiplication

Two matrices can only by multiplied together using matrix multiplication when the number of columns in the first matrix is equal to the number of rows in the second matrix

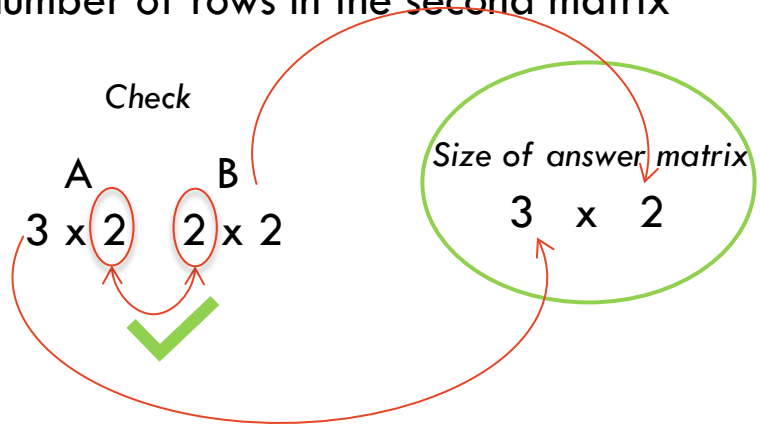
A

1	2
3	4
5	6

x

B

7	8
9	10



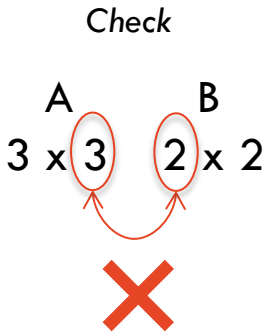
A

1	2	7
3	4	8
5	6	9

x

B

7	8
9	10



Matrix operations

Matrix multiplication

Once the dimensions have been checked, the matrix multiplication can be calculated. Mathematically this is an addition exercise (just as normal multiplication is)

3×5	$=$	$3 + 3 + 3 + 3 + 3$	$=$	15

A

1	2
3	4

x

B

5	6
7	8

$=$

1 x 5 + 2 x 7	1 x 6 + 2 x 8
3 x 5 + 4 x 7	3 x 6 + 4 x 8

$=$

19	22
43	50

Matrix operations

Matrix multiplication

Note that matrix multiplication is not commutative which means that **A** x **B** is not necessarily equal to **B** x **A**

A

1	2
3	4

x

B

5	6
7	8

=

1 x 5 + 2 x 7	1 x 6 + 2 x 8
3 x 5 + 4 x 7	3 x 6 + 4 x 8

=

19	22
43	50

≠

B

5	6
7	8

x

A

1	2
3	4

=

5 x 1 + 6 x 3	5 x 2 + 6 x 4
7 x 1 + 8 x 3	7 x 2 + 8 x 4

=

23	34
31	46

Question

Can these two matrices be multiplied together using matrix multiplication?

A

1	2
3	4
5	6

B

1	2	3	4
5	6	7	8

1 A x B?

2 B x A?

Question

A

1	2
3	4
5	6

B

1	2	3	4
5	6	7	8

1

A B

(3 x 2) x (2 x 4)



2

B A

(2 x 4) x (3 x 2)



Question

A

1	2
3	4
5	6

B

1	2	3	4
5	6	7	8

1

A B

$(3 \times 2) \times (2 \times 4)$



2

B A

$(2 \times 4) \times (3 \times 2)$



A

1	2
3	4
5	6

x

B

1	2	3	4
5	6	7	8

=

1 x 1 + 2 x 5	1 x 2 + 2 x 6	1 x 3 + 2 x 7	1 x 4 + 2 x 8
3 x 1 + 4 x 5	3 x 2 + 4 x 6	3 x 3 + 4 x 7	3 x 4 + 4 x 8
5 x 1 + 6 x 5	5 x 2 + 6 x 6	5 x 3 + 6 x 7	5 x 4 + 6 x 8

=

11	14	17	20
23	30	37	44
35	46	57	68

3 x 4

Size of new matrix

Matrix multiplication practice

4	6	3
5	8	-2

x

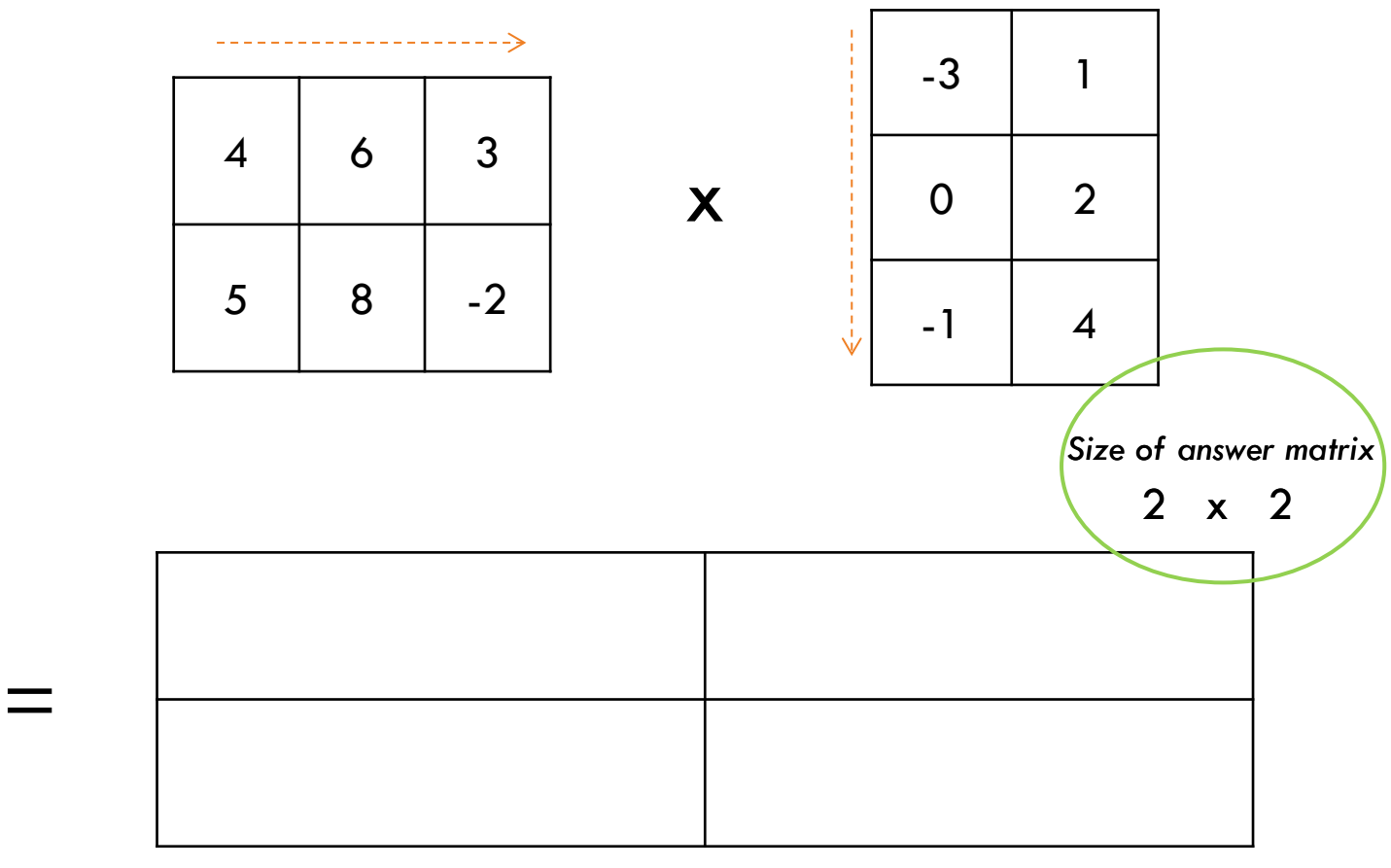
-3	1
0	2
-1	4

(2 x 3) x (3 x 2)

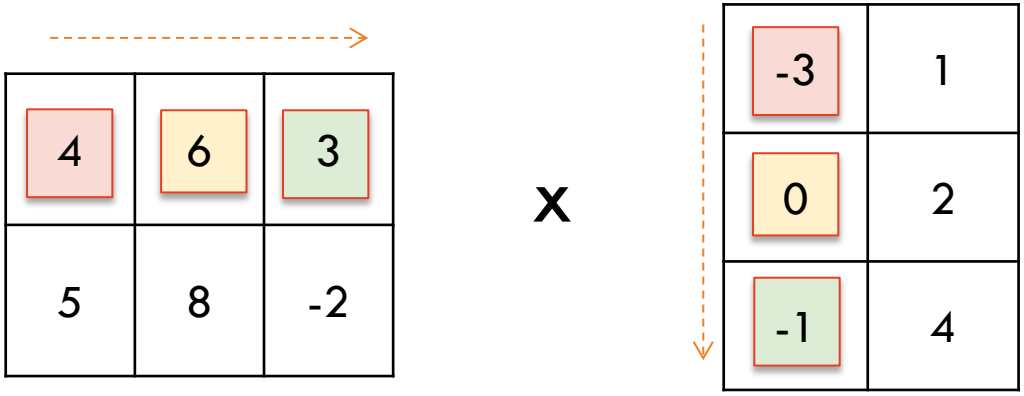
Size of answer matrix

2 x 2

Matrix multiplication practice



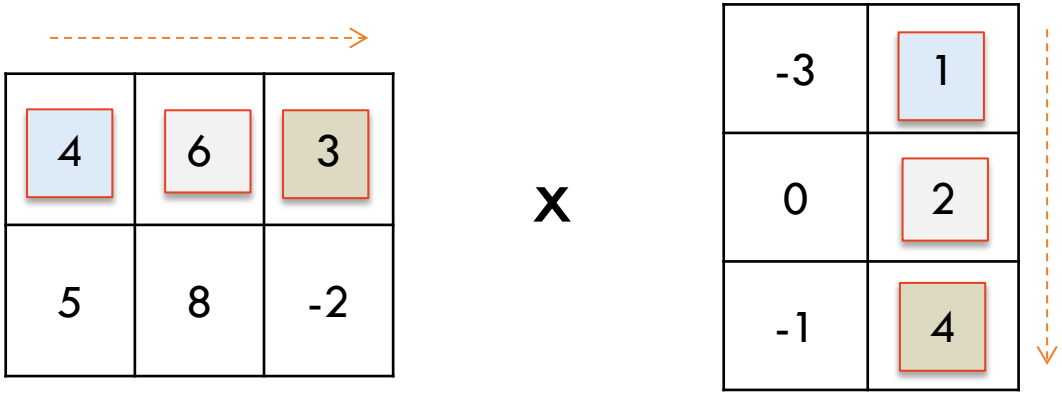
Matrix multiplication practice



=

(4x-3)+(6x0)+(3x-1)	(4x1)+(6x2)+(3x4)
(5x-3)+(8x0)+(-2x-1)	(5x1)+(8x2)+(-2x4)

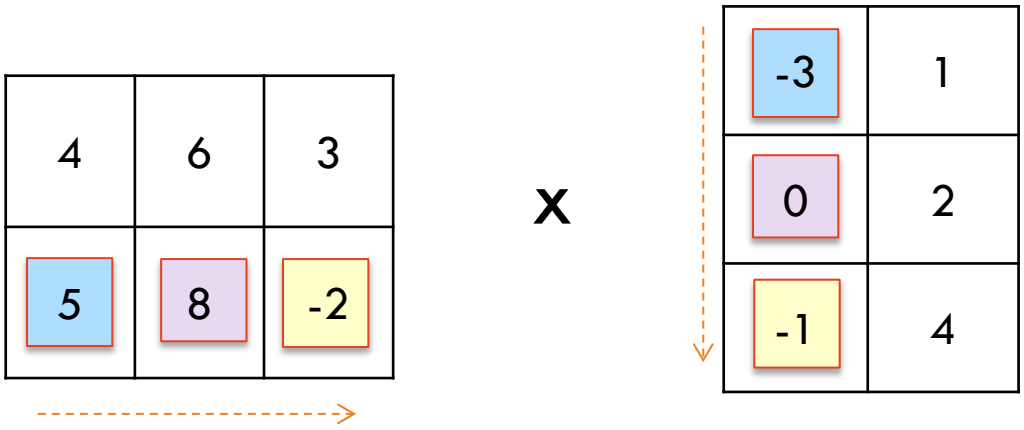
Matrix multiplication practice



=

(4x-3)+(6x0)+(3x-1)	(4x1)+(6x2)+(3x4)
(5x-3)+(8x0)+(-2x-1)	(5x1)+(8x2)+(-2x4)

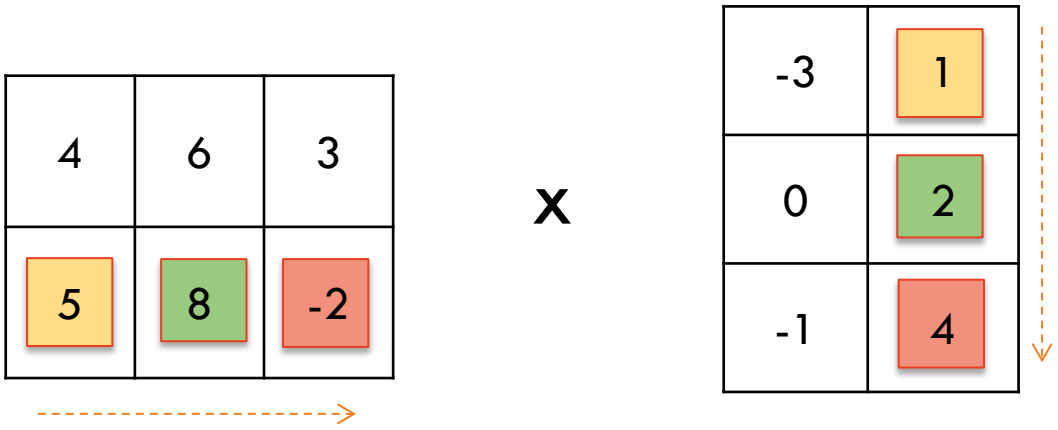
Matrix multiplication practice



=

(4x-3)+(6x0)+(3x-1)	(4x1)+(6x2)+(3x4)
(5x-3)+(8x0)+(-2x-1)	(5x1)+(8x2)+(-2x4)

Matrix multiplication practice



=

(4x-3)+(6x0)+(3x-1)	(4x1)+(6x2)+(3x4)
(5x-3)+(8x0)+(-2x-1)	(5x1)+(8x2)+(-2x4)

Matrix multiplication practice

4	6	3
5	8	-2

x

-3	1
0	2
-1	4

=

(4x-3)+(6x0)+(3x-1)	(4x1)+(6x2)+(3x4)
(5x-3)+(8x0)+(-2x-1)	(5x1)+(8x2)+(-2x4)

=

(-12 + 0 - 3) = -15	(4 + 12 + 12) = 28
(-15 + 0 +2) = -13	(5 + 16 – 8) = 13

Matrix multiplication practice – one more !

1	2	3
4	5	6

x

7	8
9	10
11	12

=

(1x7)+(2x9)+(3x11)	(1x8)+(2x10)+(3x12)
(4x7)+(5x9)+(6x11)	(4x8)+(5x10)+(6x12)

=

58	64
139	154

Matrix summation operators

Matrix summation operators allow us to convert a matrix into a single row, or a single column. This is a more efficient way of summing the elements of a row or column once the matrix becomes large.

Column summation operator

Sums down the columns to turn a matrix into a single row

$\mathbf{1}_{1x3}$

1	1	1

Row summation operator

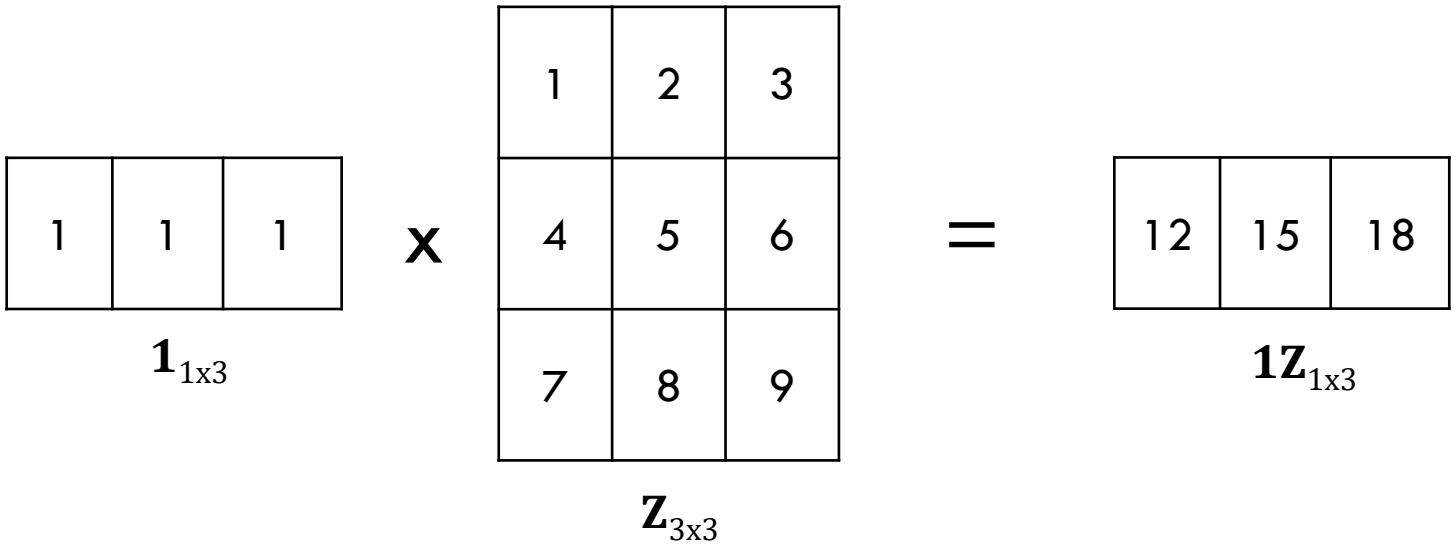
Sums across the rows to turn a matrix into a single column

$\mathbf{1}_{3x1}$

1
1
1

Column summation operator

$\mathbf{1}_{1x3}$ | 1 | 1 | 1 |



$\mathbf{1}_{1x3}$ x $\mathbf{Z}_{3x3}$ = $\mathbf{1Z}_{1x3}$

Note the order of multiplication!

Row summation operator

$\mathbf{1}_{3x1}$

1
1
1

1	2	3
4	5	6
7	8	9

$\mathbf{Z}_{3x3}$

x

1
1
1

$\mathbf{1}_{3x1}$

=

6
15
24

$\mathbf{Z1}_{3x1}$

Note the order of multiplication!

Matrix calculations in Excel

- Open Excel
- Type up the following two matrices

1	2	3
4	5	6

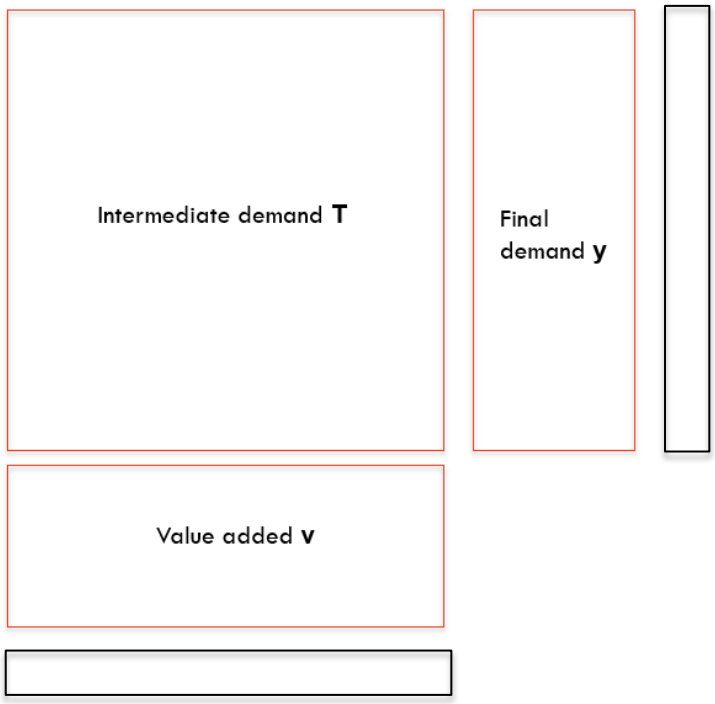
7	8
9	10
11	12

- Determine the size of the two matrices; and the size of the resultant matrix (i.e. what will the size of your final answer be if you undertake matrix multiplication?)
- Select blank cells according to the size of the resultant matrix
- Use **mmult** to multiply the 2 matrices
- Press Ctrl + Shift + Enter

Excel Help: https://support.microsoft.com/en-us/office/mmult-function-40593ed7-a3cd-4b6b-b9a3-e4ad3c7245eb

Putting matrix multiplication to work:
How do we calculate total output and total input for an input-output table?

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1 column – total output **X**

1 row – total input **X'**

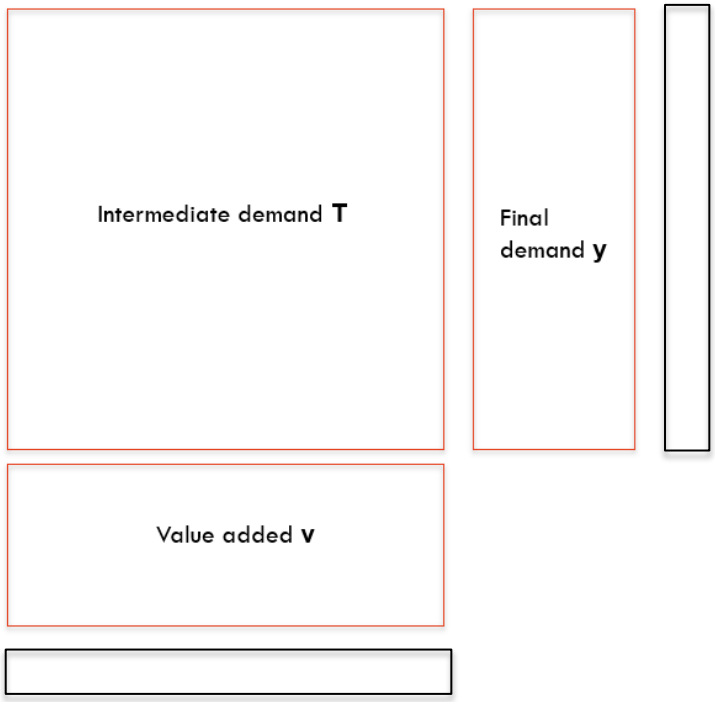
Balancing rule for input-output tables

For any particular sector: total input must equal total output

Total Inputs (X') = Total Outputs (X)

T + V = T + Y

Inputs to production Outputs to consumption



1 column – total output **x**

1 row – total input **x'**

How do you add matrices? **T** + **V**? **T** + **Y**?

Check the size of th atrices !

Total Input = **T** + **V**

Total Output = **T** + **Y**

T

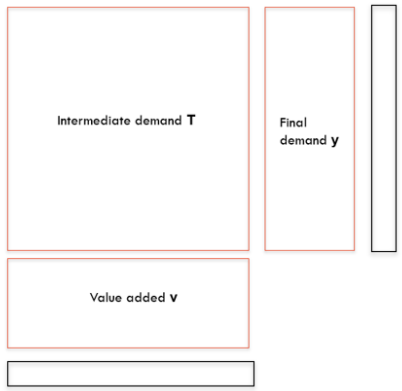
n x n

Problem!!!

Matrix operations

Addition and subtraction

In order to add or subtract matrices, the matrices in question MUST be the same size



T (4 x 4)

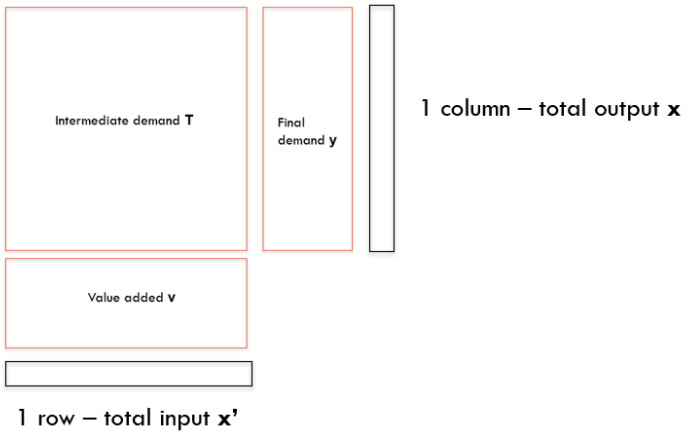
5	6	7	8
9	2	2	3
4	5	6	7
8	6	6	2

Y (4 x 3)

2	2	1
1	0	5
1	2	0
3	1	2

V (3 x 4)

2	1	1	2
2	1	1	3
1	1	2	3



Total Output = T + Y

T (4 x 4)

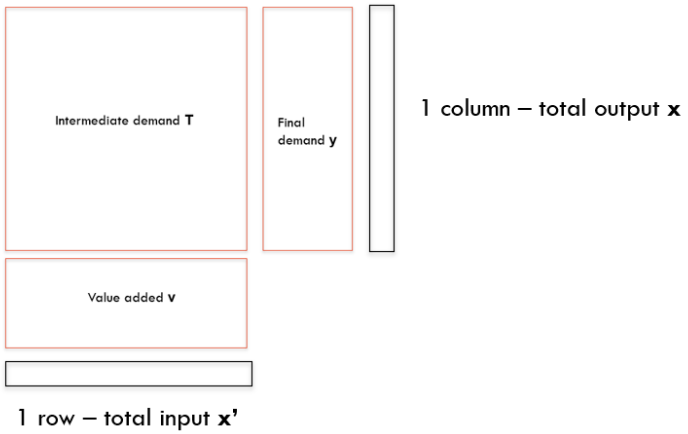
5	6	7	8
9	2	2	3
4	5	6	7
8	6	6	2

+

Y (4 x 3)

2	2	1
1	0	5
1	2	0
3	1	2

= **X** (4 x 1)



Total Output = T + Y

Solution: summation operators

$$X = T1^T + Y1^Y$$

Row summation operator for T

Row summation operator for Y

(the equivalent of multiplying by 1)

T (4 x 4)

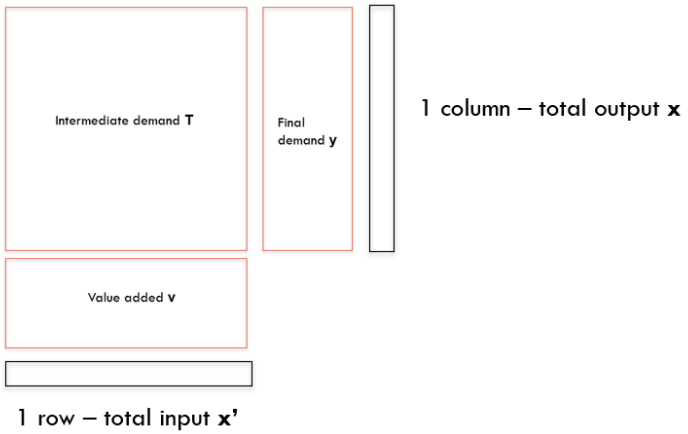
5	6	7	8
9	2	2	3
4	5	6	7
8	6	6	2

+

Y (4 x 3)

2	2	1
1	0	5
1	2	0
3	1	2

= **X** (4 x 1)



Total Output = T + Y

$$X = T1^T + Y1^Y$$

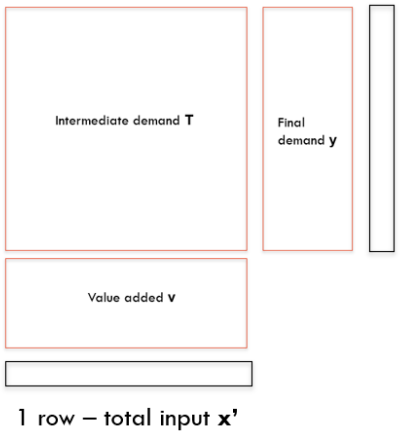
T (4 x 4)					1^T (4 x 1)	$T1^T$
5	6	7	8	x	1	26
9	2	2	3		1	16
4	5	6	7		1	22
8	6	6	2		1	22

Size check

(4 x 4) x (4 x 1)

answer will be:
(4 x 1)

Row summation operator (ie summing across the rows)



Total Output = T + Y

$$X = T1^T + Y1^Y$$

Y (4 x 3)

2	2	1
1	0	5
1	2	0
3	1	2

x

1^Y (3 x 1)

1
1
1

=

$Y1^Y$

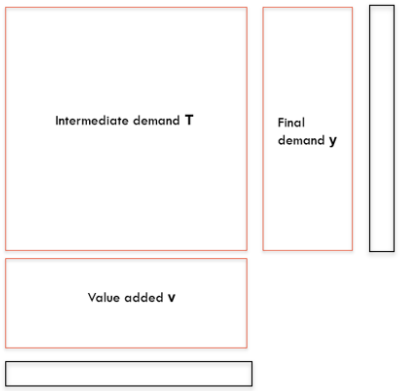
5
6
3
6

Size check

(4 x 3) x (3 x 1)

answer will be:
(4 x 1)

Row summation operator (ie summing across the rows)



Total Output = T + Y

$$X = T1^T + Y1^Y$$

T

5	6	7	8
9	2	2	3
4	5	6	7
8	6	6	2

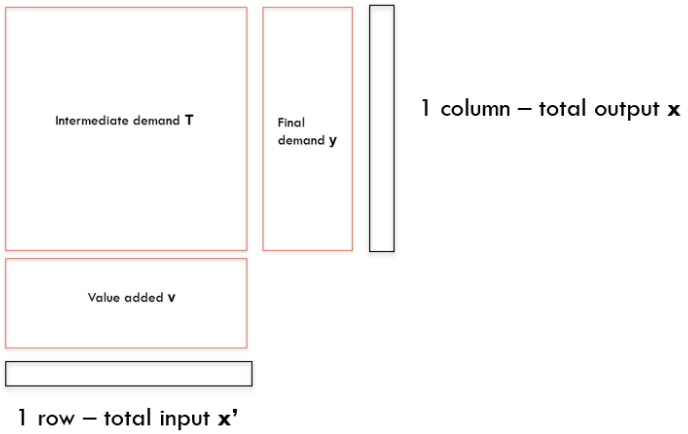
+

Y

2	2	1
1	0	5
1	2	0
3	1	2

becomes

26	+	5	=	31
16		6		22
22		3		25
22		6		28



Total Output = T + Y

$$X = T1^T + Y1^Y$$

T

5	6	7	8
9	2	2	3
4	5	6	7
8	6	6	2

+

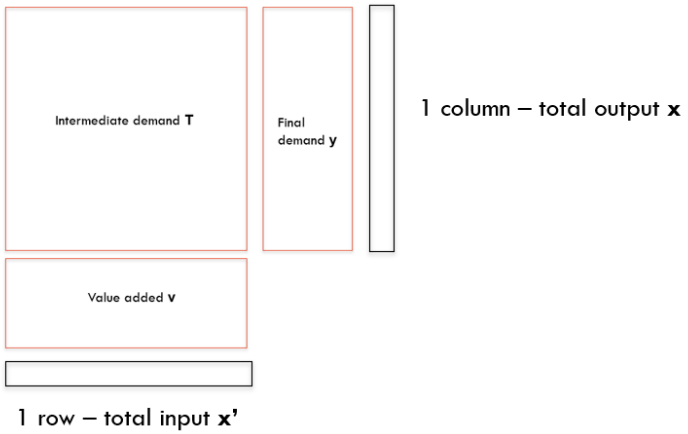
Y

2	2	1
1	0	5
1	2	0
3	1	2

=

X

31
22
25
28



T (4 x 4)

5	6	7	8
9	2	2	3
4	5	6	7
8	6	6	2

+

v (3 x 4)

2	1	1	2
2	1	1	3
1	1	2	3

= **x'**

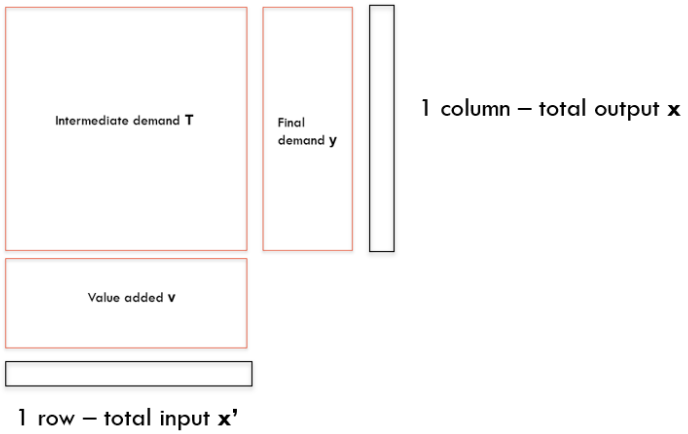
Total Input = **T** + **V**

$\mathbf{x'} = \mathbf{1}^{T}\mathbf{T} + \mathbf{1}^{v}\mathbf{y}$

(the equivalent of multiplying by 1)

Column summation operator for T

Column Summation operator for v



Column summation operator (ie summing down the columns)

$\mathbf{1}^{T'}$ (1 x 4)

1	1	1	1

X

T (4 x 4)

5	6	7	8
9	2	2	3
4	5	6	7
8	6	6	2

=

(1 x 4)

26	19	21	20

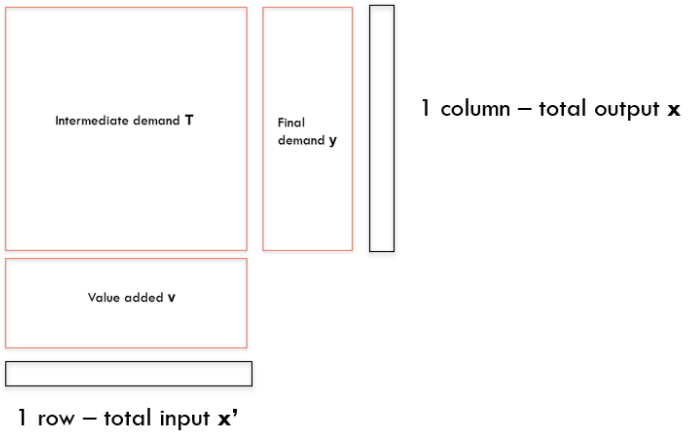
Total Input = **T** + **V**

$\mathbf{x'} = \mathbf{1}^{T'}\mathbf{T} + \mathbf{1}^{v}\mathbf{v}$

Size check

(1 x 4) x (4 x 4)

answer will be:
(1 x 4)



Column summation operator (ie summing down the columns)

Total Input = **T** + **V**

$\mathbf{1}^{V}$ (1 x 3)

1	1	1

X

V (3 x 4)

2	1	1	2
2	1	1	3
1	1	2	3

=

(1 x 4)

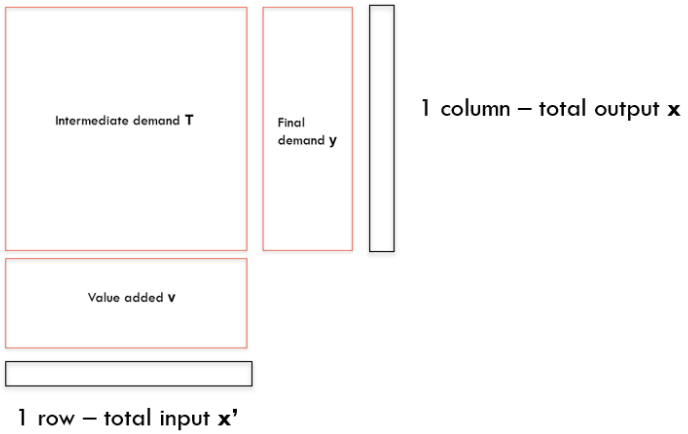
5	3	4	8

$\mathbf{x'} = \mathbf{1}^{T'}\mathbf{T} + \mathbf{1}^{v}\mathbf{v}$

Size check

(1 x 3) x (3 x 4)

answer will be:
(1 x 4)



Total Input = T + V

$$x' = 1^{T'}T + 1^{v}v$$

T (4 x 4)

5	6	7	8
9	2	2	3
4	5	6	7
8	6	6	2

+

V (3 x 4)

2	1	1	2
2	1	1	3
1	1	2	3

becomes

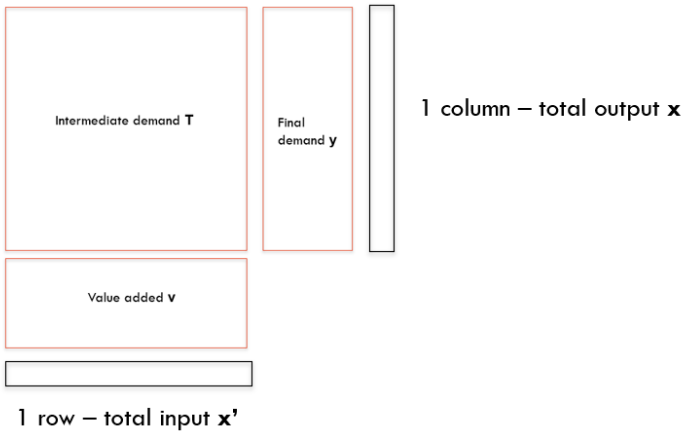
26	19	21	20

+

5	3	4	8

=

31	22	25	28



T (4 x 4)

5	6	7	8
9	2	2	3
4	5	6	7
8	6	6	2

+

V (3 x 4)

2	1	1	2
2	1	1	3
1	1	2	3

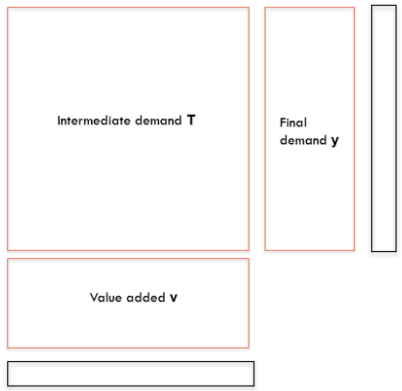
Total Input = **T** + **V**

$\mathbf{x'} = \mathbf{1}^{T'}\mathbf{T} + \mathbf{1}^{v}\mathbf{v}$

=

31	22	25	28

x' (1 x 4)



5	6	7	8
9	2	2	3
4	5	6	7
8	6	6	2

2	2	1
1	0	5
1	2	0
3	1	2

31
22
25
28

2	1	1	2
2	1	1	3
1	1	2	3

31	22	25	28

Total output for each sector

=

Total input for each sector

Why summation operators?

MATLAB !
MORE IN WEEKS
12 AND 13

T

4x4 double

	1	2	3	4
1	0.0975	0.9649	0.4854	0.9157
2	0.2785	0.1576	0.8003	0.7922
3	0.5469	0.9706	0.1419	0.9595
4	0.9575	0.9572	0.4218	0.6557
5				
6				

y

4x3 double

	1	2	3
1	0.0357	0.7577	0.1712
2	0.8491	0.7431	0.7060
3	0.9340	0.3922	0.0318
4	0.6787	0.6555	0.2769
5			
6			

```
>> x = T + y
Matrix dimensions must agree.

>> x = sum(T + y)
Matrix dimensions must agree.

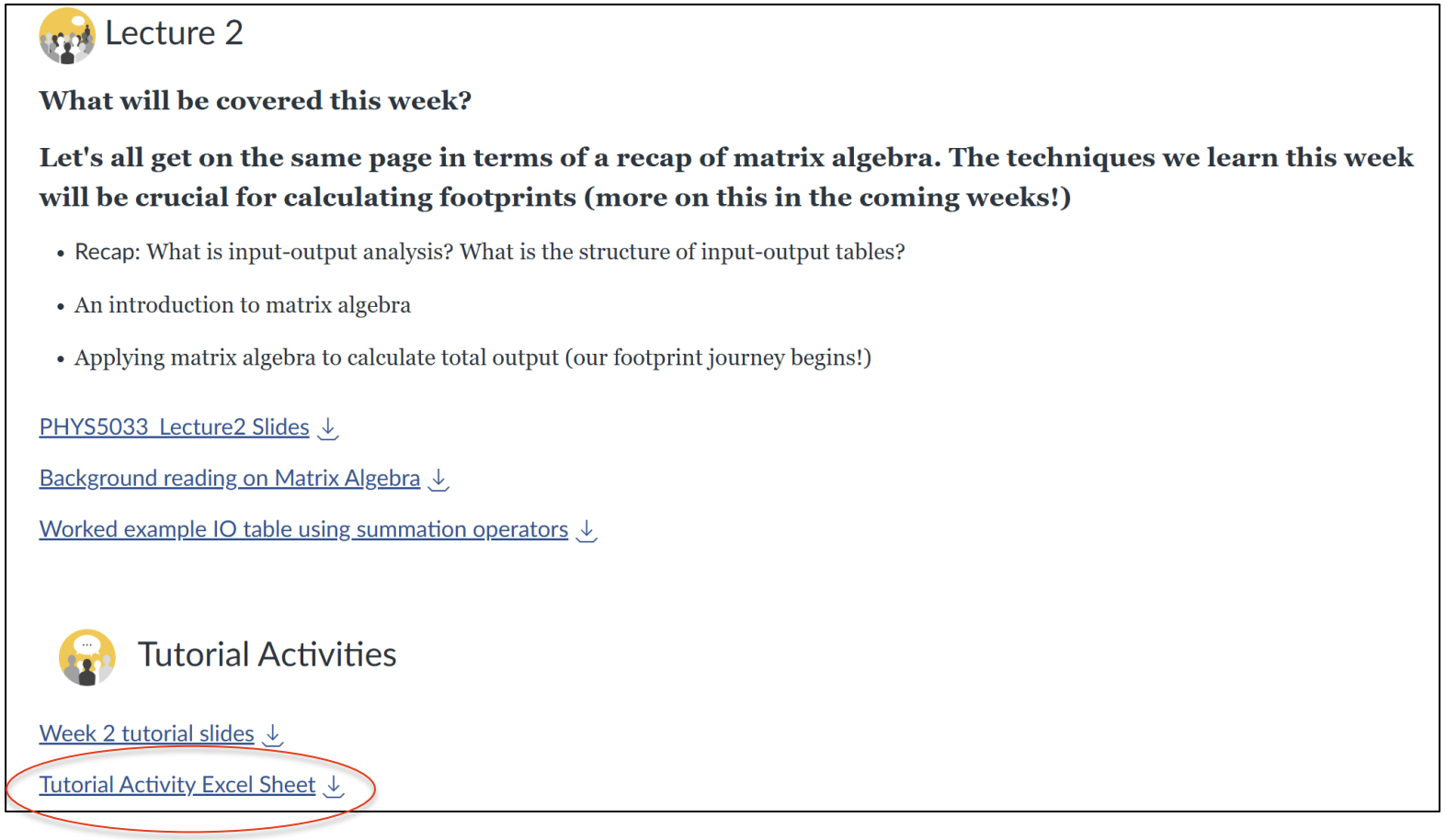
>> x = sum(T,2) + sum(y,2)

x =

    3.4282
    4.3269
    3.9769
    4.6033
```

Tutorial

Download the Tutorial Activity Excel Sheet, and work through questions 1-5 in small groups



Homework for Week 2

Homework week 2 (Due: September 9, 2026)

Matrix operations and balancing of your input-output table

In week 1 you were asked to build an input-output table. For this homework, you will be doing the following tasks using that input-output table:

- Create a new sheet in your Excel file and copy your 14-sector input-output table into this new Excel sheet.
- Add row and column summation vectors that will be used to calculate the row sum and column sum of each of the matrices in your input-output system.
- Use the matrix multiplication techniques taught in the lecture and the row and column summation vectors to calculate your total output and total input vector in the Excel sheet.
- Is your table balanced? If not, alter the values in your input-output system to balance the table.

EXPLANATION: Explain the concept of balancing for input-output tables (about 150 words).

PHYS5033 organisation – assessment tasks

Tips for the Homework tasks:

- Make sure that you read the homework instructions thoroughly for each week – if you don't complete one of the steps expected, you won't get any marks for it.
- If there is a question posed in one of the homework steps, make sure you provide an explicit answer for that question in the Excel file.
- Always include labels and units whenever you present results and make sure that your workings are included.
- You will be provided with worked examples and tutorial answers. These provide you with examples of how the calculations work – they do not provide you with a template for the homework. If you just replicate these, you will not be answering the steps expected in the homework.
- You should be able to answer the questions posed in the 'Explanation' part of the homework each week using the content provided in the lectures and in the background reading material. There is no need to use other sources to answer these questions, and doing so often results in students providing irrelevant or incorrect answers. If you do use other sources of information, these sources must be cited and referenced appropriately. You are strongly advised NOT to rely on AI tools to source your answers for these questions since they usually provide incorrect information.
- Make sure that your Excel submission is easy for the marker to read and find your results. This includes using appropriate font sizes, using appropriate row heights and column widths, and including text to note which step the results you're presenting relate to.

Key takeaways

- Matrix notation
- Mathematical operations using matrices
- The importance of checking the size of matrices
- Row and column summation operators
- Using summation operators to check if an IO table is balanced

More to come:

- How IO tables are developed
- Extracting information on economic activity from the IO table using matrix operations